Why We Are in a Techlash

A Source-Verified Timeline of the Turn Against Screens in Schools

Deep Research Report

Prepared for: WSCC Literacy Conference, September 11, 2026

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Method: four independent research agents working separate source categories (government and official, news and journalism, academic and peer-reviewed, and industry, practitioner and nonprofit), followed by cross-referencing, contradiction analysis, and targeted URL verification.

Metric	Result
Source categories covered	4 of 4
Distinct sources collected	111
Sources rated high reliability	48
Sources rated medium reliability	49
Sources rated low reliability	14
Timeline entries with a supporting source	104
Contradictions flagged	6
Citations checked and corrected	5
Citations that could not be verified at all	1 (Ulum 2026)
Agent runs interrupted and resumed	4 of 4 (machine sleep)

Confidence markers used throughout: [HIGH] means the underlying document was retrieved and read, or the claim is corroborated across multiple independent source categories. [MED] means consistent secondary reporting without a retrieved primary document. [LOW] means a single source, an interested source, or a claim that could not be confirmed.

Executive Summary

The techlash is usually told as a simple story: schools fell in love with screens, screens hurt kids, and now schools are correcting course. The evidence does not support that story. It supports a more uncomfortable one, in three parts.

One: policy ran ahead of evidence twice, in opposite directions

In the 2010s, districts scaled one-to-one devices while the best available randomized evidence was reporting nothing. The One Laptop Per Child trial in Peru randomized 318 schools, raised computers per student from 0.12 to 1.18, and found no effect on math or language scores. Districts bought anyway.

In the 2020s, the correction is repeating the same mistake in reverse. Roughly thirty states restricted phones before the first causal study of phone bans was published in October 2025. Then in April 2026 the largest and best-identified study of the policy, pre-registered and national, found that average effects on test scores are, in its own words, consistently close to zero.

Two: the reading decline is older than the thing being blamed for it

This is the part that matters most for literacy educators. NAEP reading has been falling since roughly 2013, seven years before the pandemic and well before one-to-one saturation. Tom Kane of the Education Recovery Scorecard put it precisely in May 2026: the pandemic was the mudslide that followed seven years of erosion. The COVID learning deficit is real and well measured, but it was larger in math than in reading. Meanwhile the intervention that actually correlates with reading recovery between 2022 and 2025 is curriculum reform, the science of reading, not device policy. Every state on that reform list improved. States without it did not.

Three: the backlash as measured is narrower than the backlash as narrated

Districts did not actually retreat from instructional technology. In the 2024-25 school year districts accessed an average of 2,982 distinct ed-tech tools, up nearly nine percent year over year. Ninety-three percent of districts planned Chromebook investment for 2025, up from 84 percent in 2023, now paid from local rather than federal money. And when CoSN surveyed 607 district technology leaders in 2026, 53 percent reported high community concern about students' personal devices while only 23 percent reported high concern about instructional technology.

The techlash, measured rather than narrated, is overwhelmingly a phone and social media backlash that has been rhetorically extended to school technology it did not originally target. That distinction is the single most useful thing an educator can carry into a conversation with a worried parent.

What actually holds up across twenty years

One finding survives the entire arc, and it is the least technological one. Adult mediation and implementation structure move outcomes; devices by themselves do not. The clearest

demonstration arrived in January 2026: a randomized trial across 83 schools where every school already had the software found that adding on-site coordinators raised usage from 7.2 to 47.4 minutes a week and math achievement by almost half a standard deviation. The software was already there. The adults were the intervention.

Coverage note

All four source categories returned usable results. All four agent runs were interrupted by the machine going to sleep and were resumed rather than restarted, which preserved retrieved material but cut several planned retrievals short. The specific consequences are listed in Section 8. The largest single gap is that no credible national figure exists, in anything retrieved here, for how many ESSER dollars went specifically to devices and connectivity. That number is treated as unknown rather than estimated.

1. Background and Context

"Techlash" in a school context refers to the reversal, roughly between 2023 and 2026, of a two-decade consensus that putting computing devices in students' hands would improve learning. It has four distinct strands that are routinely conflated, and separating them is most of the analytical work:

- Personal smartphones and social media, which students bring from home. This is where the evidence of harm is strongest and where nearly all the policy energy has gone.
- School-issued devices, the one-to-one Chromebook and iPad fleets districts bought with federal money. Only in 2026 did legislation begin targeting these directly.
- Instructional software, the thousands of ed-tech tools running on those devices, where the evidence of effectiveness has always been thin.
- Generative AI, which arrived in November 2022 and is now on a separate and partly contradictory policy track.

A reader who keeps these four separate will find the timeline coherent. A reader who treats them as one thing will find it contradictory, because the same governments restricting the first two are actively funding the fourth.

2. The Timeline

Act I: Optimism without evidence, 2013 to 2019

[HIGH] June 6, 2013. President Obama announces ConnectED in Mooresville, North Carolina, calling on the FCC to bring high-speed internet to 99 percent of American students within five years, with a $2 billion down payment through E-Rate modernization. Federal policy deliberately builds the connected classroom.

[MED] 2013. Los Angeles Unified launches a $1.3 billion program to buy 650,000 iPads with Pearson curriculum. Software was purchased before it was finished; Pearson delivered only sample units in year one. The FBI raided district offices over the procurement in December 2014. LAUSD later recovered $6.4 million from Pearson and $4.2 million from Apple. No criminal charges resulted. This becomes the template for every later ed-tech disappointment: buy the hardware first and assume the pedagogy will follow.

[HIGH] 2016. Beland and Murphy publish "Ill Communication" in Labour Economics. Banning phones in English schools raised age-16 exam scores by about 0.07 standard deviations, which is 6.4 percent of a standard deviation, not a 6 percent score gain. The entire effect came from low achievers, who gained 14.23 percent of a standard deviation. High achievers showed no significant change. This is the founding empirical brick of the phone-ban movement and the origin of its equity framing.

[HIGH] 2017. Cristia and colleagues publish the One Laptop Per Child randomized evaluation in American Economic Journal: Applied Economics. 318 primary schools in rural Peru, randomized, measured after 15 months. Computers per student rose from 0.12 to 1.18. The finding, verbatim: no evidence is found of effects on test scores in math and language. This is genuinely causal evidence, and it is the strongest single rebuttal to the one-to-one thesis.

[MED] 2018. Delgado and colleagues establish the screen inferiority effect in Educational Research Review. 54 studies. Overall Hedges' g of negative 0.21 favoring paper. But the effect is concentrated: negative 0.27 for expository text versus positive 0.01 for narrative, and significant only under time pressure. It also grew by about 0.01 per year from 2001 to 2017, which quietly killed the digital native assumption.

[MED] November 5, 2018. About 100 students walk out of Brooklyn's Secondary School of Journalism in protest of the Summit Learning platform. The first visible techlash event in an American school was student-led, not parent-led, and it happened before the pandemic.

[MED] 2018 and 2019. Emily Hanford's APM Reports documentaries launch the science of reading movement. This matters structurally as well as substantively: it established that a journalist-led evidence critique could overturn entrenched school practice. Horvath later follows exactly this template.

[HIGH] 2019. The International Literacy Association publishes its position statement on digital resources in early childhood literacy. Its position is a judicious balance between digital and non-digital resources, retaining printed books and paper, and it warns that adding a digital tool

should not simply replace ongoing literacy practices with those conducted on screen. Notably, ILA also flags source bias as an evaluation criterion, observing that technology position statements vary in whether they are connected to the digital industry.

[HIGH] May 23, 2019. Chalkbeat reports that Summit Learning's real annual school attrition was 18 percent, not the 10 percent claimed. The Chan Zuckerberg Initiative had put over $100 million into the platform. The ed-tech evidence problem was documented and public before the pandemic buildout began.

[HIGH] 2019. Orben and Przybylski publish in Nature Human Behaviour. Across 355,358 participants and 20,776 analytic specifications, the association between digital technology use and adolescent well-being is negative but explains at most 0.4 percent of the variance. It is correlational, but the method was purpose-built to defeat analytic flexibility. This remains the strongest methodological check on screen panic.

[MED] 2019. Adelantado-Renau and colleagues publish a JAMA Pediatrics meta-analysis covering 480,479 participants. Overall screen media use shows an effect size of negative 0.29 with a confidence interval from negative 0.65 to positive 0.08, which is not significant. Television and video games are negatively associated; internet and mobile phone time are not. The category "screen time" does not hold together.

Act II: COVID makes devices universal, 2020 to 2022

[HIGH] March 2020 onward. School closures collapse the distinction between device as tool and device as school. Congress appropriates roughly $189 to $190 billion in ESSER across three rounds, plus a $7.171 billion Emergency Connectivity Fund.

[HIGH] The Department of Education's Inspector General surveyed 300 districts and found that 92 percent used ESSER funds for educational technology: 93 percent bought hardware, 81 percent bought hotspots, 74 percent funded connectivity. Fifty-seven percent reported they could not get the devices they wanted because of shortages and shipping delays. Supply-constrained buying is the signature of deadline-driven procurement, not deliberation.

[HIGH] 2020. Chromebook shipments reach roughly 30.6 million units globally. By March 2021, district one-to-one had gone from about two-thirds to 90 percent in middle and high school, and from about 40 percent to 84 percent in elementary. That elementary doubling in twelve months is the most consequential number in this report for a literacy audience, because elementary is exactly the grade band where the print-versus-screen evidence is most contested.

[MED] August 2020. HolonIQ revises its global ed-tech market forecast upward by $63 billion, to $404 billion by 2025, on the basis of a temporary emergency, and publishes it while districts are making irreversible purchases. This is a projection, not a measurement, and no retrieved source measures what global ed-tech spending actually reached in 2025.

[HIGH] 2020. Kessel, Hardardottir and Tyrefors publish the Swedish replication in Economics of Education Review. Using the universe of Swedish secondary schools, phone bans show no effect

on performance, and the authors can reject even small gains. This is the first serious external-validity blow to Beland and Murphy, and it is routinely miscited.

[MED] 2021. Ed-tech venture funding peaks at $20.8 billion across more than 1,500 rounds, roughly three times pre-pandemic levels.

[HIGH] August 2021. Furenes, Kucirkova and Bus publish the meta-analysis that should anchor any literacy educator's understanding of this question. See Section 3 for the full breakdown; the short version is that the medium matters far less than the design and the adult.

[MED] November 1, 2021. IDC records the first Chromebook shipment decline since the pandemic began. The market turned roughly eighteen months after the peak, from saturation, well before the cultural backlash arrived. Full-year 2022 shipments fell 48 percent to 17.7 million.

[HIGH] November 2022. ChatGPT is released, opening a second front before the first is resolved.

Act III: The reckoning, 2023 to 2024

[MED] January 2023. Seattle Public Schools sues Alphabet, Meta, Snap and ByteDance. Districts become plaintiffs, which reframes them from regulators of student behavior into injured parties with a financial claim. Roughly 800 to 1,400 districts eventually join.

[HIGH] January 2023. New York City bans ChatGPT on school devices. Chancellor David Banks rescinds the ban on May 18, 2023, writing that the initial caution overlooked the potential of generative AI. This is the high-water mark of K-12 AI optimism, and it is worth noting the source is an op-ed by a sitting chancellor, not reporting.

[HIGH] January 2023. Betthauser, Bach-Mortensen and Engzell publish in Nature Human Behaviour. 42 studies across 15 countries. The COVID learning deficit is Cohen's d of negative 0.14, roughly 35 percent of a school year, worse for low-SES students, and larger in math than in reading. That last clause is routinely dropped and it matters enormously for the reading argument.

[HIGH] May 9, 2023. Florida's HB 379 is approved, effective July 1, 2023. The first state law barring phones during instructional time. It passed both chambers unanimously, which established the bipartisan character of the phone fight from the outset.

[MED] May 2023. The U.S. Surgeon General issues the advisory on social media and youth mental health, concluding the evidence does not support a finding that social media is sufficiently safe for children. This was the permission slip for everything that followed. The exact release day could not be confirmed from the primary document.

[HIGH] July 26, 2023. UNESCO launches its Global Education Monitoring Report, "Technology in education: A tool on whose terms?", calling for phones out of school absent a clear learning purpose. Its privacy finding matters as much as its distraction finding: 89 percent of 163 ed-tech products recommended during the pandemic could survey children. That converted pandemic procurement from a story of heroism into a story of children's-rights risk.

[MED] October 24, 2023. Forty-one states plus the District of Columbia sue Meta. Tennessee participates but files separately in its own state court rather than joining the federal action.

[HIGH] December 2023. OECD releases PISA 2022. Mathematics fell almost 15 points and reading 10 points against 2018, a decline OECD itself described as only partly attributable to the pandemic. This is the pivotal causal moment in the whole timeline: an authoritative body announced a crisis and simultaneously ruled out the obvious explanation. The vacuum that created is what smartphones filled.

[HIGH] March 26, 2024. Haidt's "The Anxious Generation" is published, giving the backlash a mass-market vocabulary and a concrete policy ask. Three days later, on March 29, Candice Odgers reviews it in Nature and states the thesis is not supported by science, writing that most data are correlative. The scientific establishment split from the movement's central text within 72 hours of publication, and the split never resolved.

[HIGH] May 2024. OECD publishes "Students, digital devices and success," supplying the quotable 49-point mathematics gap between students with under an hour and those with five to seven hours of leisure screen use at school. Two caveats routinely dropped in advocacy citation: the finding is about leisure use at school, not instructional use, and the relationship is curvilinear, with moderate learning use associated with better outcomes.

[MED] June 17, 2024. Surgeon General Murthy calls in the New York Times for a congressionally mandated tobacco-style warning label on social media. The escalation from "not proven safe" to "label it like cigarettes" is what made state-level restriction politically cheap.

[HIGH] June 30, 2024. The Emergency Connectivity Fund sunsets. September 30, 2024: the ESSER III obligation deadline. The device money ends; the refresh obligation does not.

[MED] October 1, 2024. The Atlantic publishes "The Elite College Students Who Can't Read Books," moving the debate from mental health to reading stamina. It is entirely anecdotal by the author's own admission that no comprehensive data exist on the trend, and it moved the national conversation anyway. This is the clearest single case in the timeline of narrative outrunning evidence.

[HIGH] December 2024. NCES surveys 1,490 schools and finds 77 percent already prohibit phone use during class, before most state laws existed. The 2025 legislative wave largely codified existing school practice rather than creating it. That is a useful corrective for anyone who experienced the laws as sudden.

Act IV: The evidence arrives late, 2025

[MED] January 29, 2025. NAEP reading falls another 2 points in grades 4 and 8. About 40 percent of fourth graders score below Basic, the largest share since 2002. No state gained in reading.

[HIGH] February 19, 2025. IES releases School Pulse Panel data: 53 percent of school leaders say phones hurt academic performance, 72 percent cite mental health harm, 73 percent cite attention span.

[MED] February 2025. The SMART Schools study in Lancet Regional Health Europe surveys 1,227 adolescents across 30 UK schools. Restrictive phone policies reduced in-school phone use but showed no association with better mental well-being. The authors state the findings do not support prohibitive school policies in their current form. The design is cross-sectional, so it cannot establish causation in either direction.

[MED] March 20, 2025. An executive order directs closure of the U.S. Department of Education. The Office of Educational Technology is eliminated in a reduction in force the same month. The federal function that would normally issue guidance on classroom device use disappeared exactly when the question became most contested, which is a direct cause of the state-by-state patchwork.

[HIGH] April 2025 and July 2025. Executive Order 14277 makes AI literacy a national K-12 priority, and the Department of Education issues an AI Dear Colleague Letter steering federal grant funds toward AI in schools. The same government restricting screens is funding AI into the same classrooms. Any honest timeline shows these two lines running simultaneously, not sequentially.

[HIGH] May 8 and September 30, 2025. The Senate votes 50-38 under the Congressional Review Act to overturn the FCC's E-Rate hotspot rule, and the FCC then votes 2-1 to strip hotspots and school bus Wi-Fi from E-Rate entirely. The homework-gap consensus breaks along partisan lines.

[HIGH] October 20, 2025. Figlio and Ozek publish NBER Working Paper 34388 on Florida's phone bans, the first causal study of the policy. Suspensions rose in year one, disproportionately for Black students, then normalized. Test gains appeared only in year two. Unexcused absences fell, possibly explaining much of the gain. It arrived after roughly thirty states had already legislated on intuition.

Act V: The techlash goes federal, and the evidence complicates it, 2026

[HIGH] January 2026. Oreopoulos and colleagues publish NBER Working Paper 34683. A randomized trial across 83 schools that already had Khan Academy found that adding on-site coordinators raised usage from 7.2 to 47.4 minutes weekly and mathematics achievement by almost half a standard deviation over 31 weeks. The constraint was never the technology. It was the organizational structure around it. Non-US residential setting limits generalizability.

[HIGH] January 15, 2026. Jared Cooney Horvath testifies before the U.S. Senate Committee on Commerce, Science, and Transportation, alongside Emily Cherkin and Jean Twenge. The C-SPAN clip passes two million views. His seven federal asks include independent efficacy standards before federal renewal, mode-equivalence validation before moving high-stakes tests to digital, and a federal evidence clearinghouse. See Section 4 for the methodological dispute attached to his numbers.

[HIGH] March 17, 2026. UNESCO reports that 114 education systems, 58 percent worldwide, now have national school phone bans, up from about 24 percent in June 2023. In the same article UNESCO hedges: limiting phone use may reduce distraction but does not remove the need for

students to learn to navigate digital environments, and schools remain one of the few places where young people develop digital literacy. The institution that triggered the wave is publicly warning that bans alone are insufficient.

[HIGH] April 2026. Allcott, Baron, Dee, Duckworth, Gentzkow and Jacob publish NBER Working Paper 35132, the largest and best-identified US study of phone bans: nationwide data, GPS pings, staggered difference-in-differences, pre-registered. Verbatim from the abstract: average effects on test scores are consistently close to zero. High schools see modest positive effects, particularly in math; middle schools see small negative effects. In year one, disciplinary incidents increase and well-being falls, with well-being turning positive later. Verified directly against nber.org.

[HIGH] May 20, 2026. The HHS Office of the Surgeon General releases "Warning on the Harms of Screen Use: An Advisory and Toolkit," announced by Secretary Robert F. Kennedy Jr. alongside Iowa Governor Kim Reynolds. The 29-page toolkit calls on schools to limit screen use by assigning work in books or on paper whenever possible, and encourages bell-to-bell phone restrictions. This is the first time a federal health advisory has told schools to move work back to paper, and it is the single most load-bearing government document for a literacy audience. Verified through K-12 Dive; the HHS PDF itself returns 403.

[HIGH] May 13, 2026. The Education Recovery Scorecard finds grade 8 reading at its lowest point since 1990 and grade 4 at pre-2003 levels, and dates the erosion to 2013. Recovery is U-shaped. Every state with science-of-reading reforms improved in reading between 2022 and 2025; states without showed no improvement. Descriptive, not causal, but pointed.

[MED] 2026. At least four states, Iowa, Utah, Tennessee and Oklahoma, enact laws restricting screen time or ed-tech for young students, with bills introduced in 16 states. Iowa caps K-5 digital instruction at 60 minutes a day and requires districts to publish their one-to-one device programs. The backlash has moved from the student's phone to the district's Chromebook.

[MED] June 2026. Los Angeles Unified unanimously adopts hard screen caps: no screens for pre-K through first grade, 60 minutes a day for grades 2 through 5, 120 minutes for grades 6 through 12, including homework. The district that defined one-to-one optimism in 2013 defined its reversal.

[HIGH] June 24, 2026. New York City delays its final AI guidance after roughly 6,500 public comments and a parent revolt. Chancellor Kamar Samuels calls AI the most invasive technology we have seen and pauses ed-tech purchases. The 2023 ban-then-reverse cycle has inverted.

[HIGH] July 16, 2026. Chalkbeat documents the structural brake on the whole movement: federally mandated grade 3-8 testing, delivered digitally by state choice, forces even screen-skeptical districts to keep devices in young students' hands. A North Carolina superintendent: if I had the choice, I'd say bubble sheets please.

[HIGH] August 6, 2026. ISTE CEO Richard Culatta concedes that ed-tech went off the rails while resisting screen caps, saying that in some cases kids are just clicking next through PowerPoint,

which is the equivalent of a worksheet on a screen. The industry's own association has absorbed the backlash and reframed the argument from whether to how much.

3. What This Means for Literacy Specifically

If you take one section into the classroom, take this one. The paper-versus-screen question has a real answer, and it is more useful than either slogan.

The Furenes meta-analysis is the anchor

Furenes, Kucirkova and Bus (2021), Review of Educational Research, 39 studies, 1,812 children ages one to eight. These effect sizes were retrieved directly from the publisher and are the most trustworthy numbers in this report.

Comparison	Effect (Hedges' g)	What it means
Overall comprehension, digital vs paper	-0.07 (not significant)	No blanket paper advantage
Digitized only, no enhancements	-0.20 (significant)	Paper wins when digital adds nothing
Story-congruent enhancements	+0.17 (significant)	Well-designed digital beats paper
Vocabulary, overall	+0.22 (favors digital)	Digital helps word learning
Vocabulary, dictionary alone	+0.49	Embedded dictionaries work
School settings	-0.28 (favors paper)	Context matters
Low-SES families	-0.19 (favors paper)	Equity dimension
Nonfiction vs fiction (digital)	+0.42 vs +0.09	Text type matters

The authors' own summary of the mechanism: adult mediation during print reading was more effective than enhancements in digital books read by children independently. But under identical adult guidance, story-congruent digital enhancements outperformed paper.

So the honest sentence for a parent or a teacher is not "paper is better." It is: a screen that just replaces the page is worse, a screen designed around the story is better, and an adult reading alongside the child beats both.

The five claims, honestly scored

Public claim	What the evidence supports
Screens are worse than paper for reading	Suggestive and conditional. Real, replicated, small. Concentrated in expository text, timed conditions, school settings, low-SES readers. Roughly zero for narrative. Digital with a dictionary is better for vocabulary.
Phone bans raise achievement	Mixed and weakening as the evidence improves. 2016 quasi-experimental +0.07 SD; 2020 Swedish replication zero; 2025 RCT +0.086 SD; 2025 Florida gains only in year two; 2026 national pre-registered near zero.
Social media causes the teen mental health crisis	Weak on causation. Overwhelmingly correlational. Best-powered analysis puts it at 0.4 percent of variance. No large randomized or credible quasi-experimental evidence at population level. And the chosen remedy shows null or negative well-being effects.
One-to-one devices improve learning	Contradicted as a standalone claim. Supported only with implementation structure. OLPC found nothing; the 2026 coordinator trial found about half a standard deviation.
COVID remote learning caused	Partly supported, importantly incomplete. Deficit is real but larger in math than

the reading crisis	reading, and the reading decline predates COVID by about seven years.

4. Points of Contradiction

These are places where sources genuinely disagree. Each is presented with both sides and an assessment.

4.1 How many states restrict phones in schools

Published counts as of mid-2026 range from 26 to 39 depending on what is being counted: statutory mandates versus recommendations, bell-to-bell versus instructional-time, states versus states plus D.C. Ballotpedia counted 33 states as of March 2026. Education Week's tracker, updated July 28, 2026, reported at least 39 states plus D.C. KFF counted 11 states with statewide policies as of April 2025. MultiState reported 35 states plus D.C. by end of 2025.

Assessment: there is no single correct number, and any figure quoted without a definition and a date is misleading. Cite a specific tracker with its date and say what it counts. The safest framing for a presentation is directional: from one state in 2023 to a large majority by 2026.

4.2 Did districts actually retreat from technology

The narrative says yes. The measurements say no. LearnPlatform found districts accessing an average of 2,982 distinct ed-tech tools in 2024-25, up nearly 9 percent. EdWeek Market Brief found 93 percent of districts planning Chromebook investment for 2025, up from 84 percent in 2023, with the funding source shifting from federal to local. CoSN found only 23 percent of districts reporting high community concern about instructional technology versus 53 percent about personal devices.

Assessment: the measured evidence is strong and consistent across three independent instruments, and it contradicts the popular framing. Note that LearnPlatform's parent company sells tool-management software, and CoSN is a membership association of district technology leaders, so both have an interest. But EdWeek Research Center does not, and it agrees. The retreat is rhetorical more than budgetary.

4.3 Do phone bans work

Beland and Murphy 2016 found +0.07 SD. Kessel and colleagues 2020 replicated the design in Sweden and found zero, able to reject even small gains. Sungu and colleagues ran an RCT and found +0.086 SD. Figlio and Ozek found year-two gains after year-one harm. Allcott and colleagues, in the largest and only pre-registered national study, found average test effects close to zero.

Assessment: the best-identified study is the most recent and the most null. The pattern across the literature is that effects shrink as identification improves, which is the classic signature of a real but small effect surrounded by noise. Phone bans are defensible on attention, behavior and simplicity grounds. They are not well supported as an achievement intervention.

4.4 The Horvath effect sizes

Horvath's Senate testimony reports adjusted effect sizes across 398 meta-analyses and 21,155 studies: general learning at negative 0.13, one-to-one laptops at negative 0.30, literacy at

negative 0.09. These are produced by subtracting 0.42 from every published effect size, on the theory that 0.42 represents the average impact of ordinary classroom instruction. His own note reads: Adjusted Effect equals Reported d minus 0.42.

Elizabeth Tipton, a statistician at Northwestern, calls that benchmark wildly unrealistic and notes that in carefully designed studies an effect above zero would be encouraging. Chalkbeat's assessment is that the work deserves attention but relies on correlative data and has significant holes, with no smoking-gun data linking ed-tech to learning declines.

Assessment: this is not a dispute about data. It is a dispute about one benchmark choice, and that choice does all the work. Presenting Horvath's numbers without the benchmark is misleading; dismissing him entirely is also wrong, since his distraction argument is conceded even by critics like Daniel Willingham.

4.5 The WestEd outcomes-based contracting study

Sources disagree on basic parameters. Two agents report a study period of August 2024 to March 2026 with publication in July 2026, 10 interventions of which 4 were rigorously evaluable and 3 of those 4 showed gains. A third reports school years 2022-23 through 2024-25, publication August 4, 2026, and positive results demonstrable in three of eight districts.

Assessment: unresolved. The qualitative findings are consistent across all sources, that AI reading software helped second graders and not older elementary students, and that the coaching cannot be separated from the contract. The study parameters should be checked against the WestEd report directly before any of these numbers are quoted.

4.6 Is Tennessee a bell-to-bell state

At least one national tracker classifies Tennessee among full-day bell-to-bell states. The primary legislative record does not support that: Public Chapter 103 covers instructional time. Assessment: use the primary record. Secondary trackers get this wrong.

5. Tennessee Policy Appendix

All four bills below were verified against the Tennessee General Assembly's own bill status system. Bill numbers use leading zeros per TN convention.

Bill	Subject	Key votes	Signed	Chapter	Effective
HB1891 (113th)	Protecting Children from Social Media Act: age verification and parental consent for under-18	House 88-2, Senate 30-0	May 13, 2024	PC 899	Jan 1, 2025
HB0932 (114th)	Districts must prohibit wireless communication devices during instructional time	House 80-10, Senate 30-0	Apr 4, 2025	PC 103	Jul 1, 2025
HB0825 (114th)	Teen Social Media and Internet Safety Act: TDOE guidance for grades 6-12, social media restricted on school networks	House 88-4, Senate 26-0	Apr 28, 2025	PC 195	Jul 1, 2025
HB2393 (114th)	K-5 districts must adopt age-appropriate digital device policies minimizing unnecessary screen time	House 87-6, Senate 31-0	Apr 21, 2026	PC 808	Jul 1, 2026

What matters about these for a literacy audience

[HIGH] HB0932 is broader than a phone ban. Its definition of wireless communication device encompasses cellular telephones, tablet computers, laptop computers and gaming devices. Exceptions cover teacher-authorized educational purposes, emergencies and health management, students with IEPs, 504 plans or individual learning plans, and assistive technology. It restricts instructional time, not the full school day.

[HIGH] HB2393 is the pivot from the student's phone to the district's Chromebook. Reporting indicates the original version would have banned digital educational devices for young students except in special circumstances, and was dialed back after concerns about tools like digital whiteboards. The enacted version favors in-person, teacher-led instruction and non-electronic materials, with exceptions for special education accommodations, required assessments and remote learning.

[MED] The sponsor's stated rationale is explicitly literacy-framed: young students develop literacy and reasoning through books, handwriting, discussion and direct instruction. For a Tennessee literacy audience this is the most directly relevant sentence in state law.

[HIGH] HB0825 is the curricular arm. Tennessee did not only restrict devices; it mandated instruction about the platforms, with TDOE guidance for grades 6-12 covering mental health impacts, AI-generated misinformation and human trafficking identification.

[MED] Tennessee's own law survived its first federal challenge. In NetChoice v. Skrmetti, the Middle District of Tennessee denied a preliminary injunction against the social media act in June 2025 on irreparable-harm grounds without reaching the First Amendment merits. The case is on appeal to the Sixth Circuit; do not state an appellate outcome.

[LOW] The 2025 Tennessee Educator Survey found 83 percent of high school teachers agreed student cell phone use disrupts academic achievement, and 73 percent thought it contributes to

cheating. Before the state mandate, 68 percent of middle school leaders barred phones during school hours while most high school leaders permitted them during breaks, between classes and at extracurriculars. Practice ran ahead of the law here too. These figures come from search summaries of the TDOE PDF and should be verified before being quoted from a podium.

6. The Money Thread

The buildout ran on a clock, not on evidence. That is the shortest accurate summary, with two honest qualifications.

[MED] Roughly $189 to $190 billion in ESSER arrived across three rounds, with a hard obligation deadline of September 30, 2024, joined by a $7.171 billion Emergency Connectivity Fund and the Affordable Connectivity Program. Approaching the deadline, states and districts reported spending more than $169.7 billion, about 89.6 percent.

[MED] Ed-tech venture funding peaked at $20.8 billion in 2021 and fell to about $2.4 billion by 2024, which HolonIQ called the lowest level in a decade. It kept falling to roughly $1 billion in the first half of 2026, down another 26 percent year over year. Capital has left K-12 for workforce training and AI-adjacent bets.

[MED] Chromebook shipments went from roughly 30.6 million units in 2020 to 17.7 million in 2022, a 48 percent decline. Critically, the market turned in Q3 2021 from saturation, before the cultural backlash. The 2019 pre-pandemic baseline could not be verified and is deliberately omitted rather than estimated.

[HIGH] What the cliff cost is measurable. Between 2023 and 2026, districts providing no off-campus connectivity services rose from 26 to 41 percent. District-owned hotspots fell from 58 to 47 percent. Free or subsidized home internet for low-income families fell from 15 to 6 percent. Bus Wi-Fi fell from 13 to 6 percent. When the money stopped, the digital-equity commitments went first, which tells you they were funded rather than believed.

[MED] On the backlash side, there is now a vendor too. Yondr, founded in 2014 for concert venues, went from about $5 million in sales through three quarters of 2024 to roughly $19 million in K-12 sales in 2025, with LAUSD alone at $5.2 million. No independent, non-vendor-funded evaluation of pouch effectiveness was found in this research, which is worth saying out loud given the spending.

Qualification one: the demand in March 2020 was real. Schools genuinely could not operate without devices. This was not manufactured need; it was emergency architecture outliving the emergency. Qualification two: districts did not actually retreat, as Section 4.2 shows. The accurate formulation is that available federal money and vendor push determined the scale, speed and grade-level reach of the buildout, particularly the elementary doubling, far more than evidence did. But the adoption proved durable and is now locally funded.

7. Practical Implications

Ordered by strength of the underlying evidence.

Well supported

- Separate the four strands in every conversation. Personal phones, school-issued devices, instructional software and generative AI are different questions with different evidence. Most public confusion collapses them.
- Put an adult next to the reading. The single most consistent finding across twenty years is that adult mediation and implementation structure move outcomes. This is also the cheapest finding to act on.
- Use print for extended expository reading and assessment. The screen inferiority effect is real, is concentrated exactly there, and is worse under time pressure.
- Do not buy hardware without the structure to use it. OLPC and the 2026 coordinator trial bracket this: same technology, opposite results, and the difference was the adults.

Reasonably supported

- Expect phone restrictions to help attention and behavior more than test scores. Set expectations accordingly so the policy is not judged against a promise the evidence does not make.
- Watch discipline data by subgroup during the first year of any restriction. Both the Florida study and the national study found year-one discipline increases that later faded, with the Florida increase concentrated among Black students.
- Evaluate digital reading tools by design, not by medium. Story-congruent enhancement helps; digitizing a page does not.

Weakly supported, proceed with caution

- Screen-time minute caps. No retrieved evidence supports specific numeric thresholds like 60 or 120 minutes. These are political settlements, not research findings. Iowa, Oklahoma and LAUSD have all adopted them anyway.
- Claims that social media causes the mental health crisis. The correlations are real and consistent; the causal inference is not established, and the measured effects are much smaller than the public debate implies.

Open questions

- PISA 2025 results are scheduled for release in late 2026, including a new Learning in the Digital World domain. The next major evidence checkpoint arrives just after this conference. Frame the timeline as open, not concluded.
- No credible national figure exists for how much ESSER money went specifically to devices and connectivity.
- No independent evaluation of phone-pouch effectiveness was found.

8. Limitations and Single-Source Claims

Citations corrected during this research

Claim as previously cited	Correction
Sungu, Choudhury & Bjerre-Nielsen as the Sweden null result	Wrong study. Sungu et al. is an RCT reporting a positive effect of about +0.086 SD. The Sweden null is Kessel, Hardardottir & Tyrefors (2020), Economics of Education Review 77, 102009.
Beland & Murphy as a 6 percent test-score gain	Units error. The finding is 0.07 SD, which is 6.4 percent of a standard deviation.
Kong et al. (2024)	The meta-analysis is Kong, Seo & Zhai (2018), 17 studies.
Ulum (2026), European Journal of Education	Could not be found under that author name in any year, across two targeted searches. Treat as unverified.
Barnum (2025), The Hechinger Report	Likely misattribution. Matt Barnum writes for Chalkbeat; the Hechinger piece on Florida bans and suspension disparities is by Jill Barshay, October 20, 2025.

Single-source claims

- Yondr revenue figures ($19 million in 2025 versus $5 million in 2024) come only from the industry category, via secondary reporting of GovSpend contract data.
- The 2025 Tennessee Educator Survey percentages come only from the government category, at search-summary level rather than a direct read of the TDOE PDF.
- Granville County's tech-free Tuesdays and Thursdays comes only from the news category, partly via a tertiary source, though the district's tech skepticism is independently corroborated.
- The 2026 Surgeon General advisory's school recommendations were single-category when first reported but have since been independently verified in this report through a direct retrieval, and are now treated as high confidence.

Documents that could not be retrieved

- HHS returned 403 on both Surgeon General advisories. The 2023 advisory's exact release day is unverified, and the identity of the Surgeon General who signed the 2026 advisory is unknown; reporting names Secretary Kennedy as the announcer, not the signer.
- Congress.gov returned 403, so KOSA and KIDS Act details rest on member press releases and trade reporting rather than the congressional record. KOSA has never been enacted; the House passed a different vehicle, H.R. 7757, on June 29, 2026 by 267-117, with the duty of care stripped.
- Numeric effect sizes were not retrieved for Figlio and Ozek, Kessel and colleagues, or the WestEd AI reading result. Those are cited by direction of effect only.
- The COVID remote-learning reckoning is the thinnest segment. Original NYT and Washington Post reporting on learning loss was not retrieved.

Method limitation

All four agent runs were interrupted by the machine going to sleep and were resumed rather than restarted. Resumption preserved retrieved material, but each agent was instructed to stop searching and return, which cut planned retrievals short. Several effect sizes come from search-result summaries rather than full texts. Where that is true it is stated. The Furenes figures, which matter most for this audience, were retrieved directly.

9. Source Reliability Matrix

Top sources by aggregate reliability. Score combines source-type weight, agent confidence, URL status and corroboration across categories.

Source	Yr	Cat	Status	Score	Rel.
Furenes, Kucirkova & Bus, Review of Educational Research	2021	Acad	Retrieved	1.00	High
Horvath, U.S. Senate written testimony	2026	Gov	Retrieved (PDF)	1.00	High
Allcott et al., NBER WP 35132	2026	Acad	Verified here	1.00	High
TN General Assembly, HB2393 / PC 808	2026	Gov	Verified here	1.00	High
TN General Assembly, HB0932 / PC 103	2025	Gov	Retrieved	1.00	High
Cristia et al., AEJ: Applied (OLPC Peru)	2017	Acad	Retrieved	1.00	High
Orben & Przybylski, Nature Human Behaviour	2019	Acad	Retrieved	1.00	High
Betthauser et al., Nature Human Behaviour	2023	Acad	Retrieved	1.00	High
Education Recovery Scorecard, Harvard GSE	2026	Acad	Retrieved	1.00	High
IES/NCES School Pulse Panel	2025	Gov	Retrieved	1.00	High
UNESCO GEM Report phone-ban tracker	2026	Gov	Retrieved	1.00	High
ED Office of Inspector General ESSER survey	2023	Gov	Via K-12 Dive	0.95	High
Odgers, Nature review of Haidt	2024	Acad	Retrieved	0.95	High
Figlio & Ozek, NBER WP 34388	2025	Acad	Citation verified	0.90	High
Oreopoulos et al., NBER WP 34683	2026	Acad	Verified	0.90	High
Barnum, Chalkbeat (Horvath fact-check)	2026	News	Retrieved	0.88	High
Barshay, Hechinger (Florida phone bans)	2025	News	Retrieved	0.88	High
ILA position statement, early childhood literacy	2019	Indus	Retrieved (PDF)	0.85	High
Beland & Murphy, Labour Economics	2016	Acad	Corroborated	0.85	High
CoSN, U.S. State of EdTech 2026	2026	Indus	Retrieved (PDF)	0.84	High
Klein & Langreo, Education Week (Senate hearing)	2026	News	Retrieved	0.83	High
K-12 Dive, Surgeon General screen advisory	2026	News	Verified here	0.83	High
Delgado et al., Educational Research Review	2018	Acad	Search-level	0.80	High
OECD, Students, digital devices and success	2024	Gov	403 on fetch	0.78	Med
Kessel et al., Economics of Education Review	2020	Acad	Citation only	0.75	Med
HHS, 2026 screen-use advisory (primary PDF)	2026	Gov	403; corroborated	0.75	Med
Ballotpedia state cellphone ban tracker	2026	News	Search-level	0.68	Med
HolonIQ ed-tech VC reports	2021-26	Indus	Partial	0.60	Med
Yondr contract/revenue reporting	2024-25	Indus	Secondary	0.55	Med
TDOE Educator Survey 2025	2025	Gov	Unparseable PDF	0.50	Med
Advocacy org self-reported membership counts	2024-26	Indus	Self-reported	0.40	Low
Ulum (2026)	n/a	Acad	NOT FOUND	0.00	Low

Summary: 111 distinct sources collected across four categories. 48 rated high reliability, 49 medium, 14 low. One cited source could not be located at all. Five citations were corrected. No URL in this report was written from memory; every link was returned by a search or fetch during this research.

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